CONTROLLING THE NUMBER OF BRANCHES OF THE FIBRES BY THE NORMALISATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10333-10352 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10354-10402, the complete author proof (47 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Normalisation of a locally Noetherian scheme, finiteness of the normalisation, and the behaviour of fibres of a finite morphism; the spreading argument is retained.

Source reference. Original lemma algebra-lemma-henselization-different: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10402–10402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma algebra-lemma-local-flat-ff: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10402–10402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma algebra-lemma-strict-henselization-different: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10402–10402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-nr-minimal-primes: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10303–10309. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-etale-at-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10402–10402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma properties-definition-number-of-branches: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10402–10402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma properties-lemma-number-of-branches-1: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10402–10402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a scheme and $x \in X$ a point. Then*

(1) *the number of branches of X at x is equal to the supremum of the number of irreducible components of U passing through u taken over elementary étale neighbourhoods $(U, u) \to (X, x)$,*
(2) *the number of geometric branches of X at x is equal to the supremum of the number of irreducible components of U passing through u taken over étale neighbourhoods $(U, u) \to (X, x)$,*
(3) *X is unibranch at x if and only if for every elementary étale neighbourhood $(U, u) \to (X, x)$ there is exactly one irreducible component of U passing through u, and*
(4) *X is geometrically unibranch at x if and only if for every étale neighbourhood $(U, u) \to (X, x)$ there is exactly one irreducible component of U passing through u.*

Proof. Parts (3) and (4) follow from parts (1) and (2) via Properties, Lemma properties-lemma-number-of-branches-1.

Proof of (1). Let $\mathrm{Spec}(A)$ be an affine open neighbourhood of x and let $\mathfrak{p} \subset A$ be the prime ideal corresponding to x. We may replace X by $\mathrm{Spec}(A)$ and it suffices to consider affine elementary étale neighbourhoods (U, u) in the supremum as they form a cofinal subsystem. Recall that the henselization $A^h_\mathfrak{p}$ is the colimit of the rings $B_\mathfrak{q}$ over the category of pairs $(B, \mathfrak{q})$ where B is an étale A-algebra and $\mathfrak{q}$ is a prime lying over $\mathfrak{p}$ with $\kappa(\mathfrak{q}) = \kappa(\mathfrak{p})$, see Algebra, Lemma algebra-lemma-henselization-different. These pairs $(B, \mathfrak{q})$ correspond exactly to the affine elementary étale neighbourhoods (U, u) by the correspondence between rings and affine schemes. Observe that irreducible components of $\mathrm{Spec}(B)$ passing through $\mathfrak{q}$ are exactly the minimal prime ideals of $B_\mathfrak{q}$. The number of minimal primes of $A^h_\mathfrak{p}$ is the number of branches of X at x by Properties, Definition properties-definition-number-of-branches. Observe that the transition maps $B_\mathfrak{q} \to B'_{\mathfrak{q}'}$ in the system are all flat. Since a flat local ring map is faithfully flat (Algebra, Lemma algebra-lemma-local-flat-ff) we see that the lemma follows from Lemma lemma-nr-minimal-primes.

Proof of (2). The proof is the same as the proof of (1), except that we use Algebra, Lemma algebra-lemma-strict-henselization-different. There is a tiny difference: given a separable algebraic closure κ^{sep} of $\kappa(x)$ for every étale neighbourhood (U, u) we can choose a $\kappa(x)$-embedding $\phi : \kappa(u) \to \kappa^{sep}$ because $\kappa(u)/\kappa(x)$ is finite separable (Morphisms, Lemma morphisms-lemma-etale-at-point). Hence we can look at the supremum over all triples (U, u, ϕ) where $(U, u) \to (X, x)$ is an affine étale neighbourhood and $\phi : \kappa(u) \to \kappa^{sep}$ is a $\kappa(x)$-embedding. These triples correspond exactly to the triples in Algebra, Lemma algebra-lemma-strict-henselization-different and the rest of the proof is exactly the same. □